

Moments / Expectations

Expectations

- ▶ We'd often like to summarize the most important features of a random variable.
- ▶ In financial econometrics, this will often be mean, variance, skewness and kurtosis.
- ▶ To understand these quantities, we must first talk about expectations.
- ▶ An expectation is the average of a random variable and summarises the central tendency of a random variable.

Expected Value: Definitions

big $X \Rightarrow$ R.V.
small $x \Rightarrow$ a number
e.g. $x=0$
 $x=1$

Discrete Random Variable

If X is a discrete random variable with **pmf** $p(x)$, then the expected value of X , denoted $\mathbb{E}[X]$, is given by

x	$p(x)$
1	0.605
0	1 - 0.605

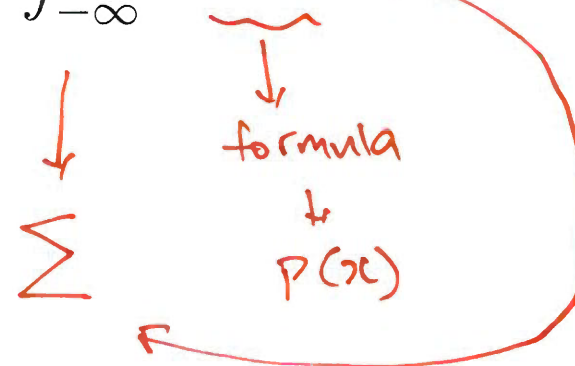
$$\mu = \mathbb{E}[X] = \sum_{x \in S_X} x \cdot \underline{p(x)}.$$

Continuous Random Variable

If X is a continuous random variable with **pdf** $f(x)$, then the expected value of X , denoted $\mathbb{E}[X]$, is given by

$x=1 \Rightarrow r > 0$

$$\mu = \mathbb{E}[X] = \int_{-\infty}^{\infty} x \cdot f(x) dx.$$



Variance and Standard Deviation

For a discrete or continuous random variable X , the variance and standard deviation of X are defined as:

$$\sigma^2 = \text{var}(X) = \mathbb{E}[(X - \mu)^2]$$

me ↑ *average* ↗
↓ *compare.*

$$\sigma = \text{sd}(X) = \sqrt{\sigma^2} = \sigma.$$

Because σ^2 represents an average squared deviation, it is not in the same units as X .

The standard deviation of X is the square root of the variance and is in the same units as X .

Example

Let $X \sim N(0, 1)$, then $\mathbb{E}[X] = 0$ and $\mathbb{V}[X] = 1$

Example

Let $X \sim N(\mu, \sigma^2)$, then $\mathbb{E}[X] = \mu$ and $\mathbb{V}[X] = \sigma^2$

Example

Let $X \sim \text{Bern}(p)$, then $\mathbb{E}[X] = p$ and $\mathbb{V}[X] = p(1 - p)$

$$\text{var}(x) = p(1-p) \quad \curvearrowright$$

Standardizing a random variable

Let X be a normal random variable with $E(X) = \mu$ and $\text{var}(X) = \sigma^2$.
That is

$$X \sim N(\mu, \sigma^2).$$

$$R \sim N(0.05, 0.1^2)$$

Define a new random variable Z as

$$Z = \frac{X - \mu}{\sigma} \sim N(0, 1). \quad \Rightarrow \quad \text{only need one table.}$$

This transformation is called standardizing the random variable X (see next slide).

Standardizing a random variable

must know.

Re-write

$$Z = \frac{X - \mu}{\sigma} = \frac{1}{\sigma}X - \frac{\mu}{\sigma}$$

and so

$$E(Z) = \frac{1}{\sigma}E(X) - \frac{\mu}{\sigma} = \frac{1}{\sigma}\mu - \frac{\mu}{\sigma} = 0.$$

$$\text{var}(Z) = \frac{1}{\sigma^2}\text{var}(X) = \frac{1}{\sigma^2}\sigma^2 = \underline{1}.$$

In short, if $X \sim N(\mu, \sigma^2)$ then $Z \sim N(0, 1)$.

Computing probabilities using standard normal

Let $X \sim N(2, 4)$ and suppose we want to find $\Pr(X > 5)$ but we only know probabilities associated with a standard normal random variable $Z \sim (0, 1)$.

We solve the problem by standardizing X as follows:

$$\Pr(X > 5) = \Pr\left(\frac{X - 2}{\sqrt{4}} > \frac{5 - 2}{\sqrt{4}}\right) = \Pr\left(Z > \frac{3}{2}\right) = \underline{\underline{0.0668}}$$

$Z = \frac{X - \mu}{\sigma} = \left(\frac{5 - 2}{\sqrt{4}}\right)$

Moments

Recall the four features of an asset return an investor would like to know:

1. The expected return from investing in an asset.
2. The risk associated with an investment.
3. The likelihood of extreme returns being above or below the mean return.
4. The likelihood of extreme returns.

Each of the above can be related to some “expectation or moment” of returns.

Moments

For each integer k , the **k -th moment** of X is

$$\mu_k = \mathbb{E}[X^k].$$

$$k=1$$

$$\mu_1 = E(X)$$

The **k -th central moment** of X is

$$\bar{\mu}_k = \mathbb{E}[(X - \mu_1)^k].$$

$$k=2$$

$$\begin{aligned}\bar{\mu}_2 &= \mathbb{E}[(X - \mu_1)^2] \\ &= \sigma^2 \Rightarrow \text{var}(X) \\ &\text{a number}\end{aligned}$$

The **k -th standardized moment** of X is given by

$$\sqrt{\bar{\mu}_2} = \sigma = \text{std dev}$$

$$\bar{\mu}_k^s = \mathbb{E}\left[\left(\frac{X - \mu_1}{\sqrt{\bar{\mu}_2}}\right)^k\right].$$

$$\bar{\mu}_k^s = E(Z^k)$$

Examples

$$\text{var}(X) = E(X - \mu)^2 = E(X^2) - E(X)^2$$

k^{th} moment
 $\downarrow$

$(E(X))^2$
 $\downarrow$
 μ^2

Example

Let $X \sim N(0, 1)$, then $E[X] = 0$ and $E[X^2] = 1 - 0^2 = 1$

Example

Let $X \sim N(\mu, \sigma^2)$, then $E[X] = \mu$ and $E[X^2] = \sigma^2 + \mu^2$

Example

Let $X \sim \text{Bern}(p)$, then $E[X] = p$ and $E[X^2] = p(1 - p) + p^2 = p$

$$\text{var}(X) = p(1-p)$$

How to estimate these Moments?

- ▶ Moments of asset returns are of critical importance for financial econometrics.
- ▶ The “expected return” can be measured by the mean: $\mathbb{E}[r_t]$.
- ▶ The “risk associated with an investment” can be measured by the variance: $\bar{\mu}_2 = \mathbb{E}[(r_t - \mu_1)^2]$.

Skewness

$$z = \frac{x - \mu}{\sigma}$$

Likelihood of extremes above or below mean: referred to as **skewness**.

Measures whether more extreme positive or negative returns are likely to occur.

Skewness is more commonly measured by

$$\bar{\mu}_3^s = \mathbb{E} \left[\left(\frac{r_t - \mu_1}{\sqrt{\bar{\mu}_2}} \right)^3 \right]. \quad \mathbb{E}(z^3)$$

1. If $\bar{\mu}_3^s < 0$, the distribution is negatively skewed.
2. For **stock returns data**, **negative skewness** is more likely than **positive skewness**, and hence large negative returns more likely than large positive returns.

compare with "0"

Kurtosis

- ▶ Likelihood of extremes: referred to as the **kurtosis**.
- ▶ This measures the likelihood of extreme events, either positive or negative.
- ▶ Extreme events can be thought of as points far away from the center of the distribution, i.e., the mean.
- ▶ Kurtosis is measured by the fourth standard moment

$$\bar{\mu}_4^s = \mathbb{E} \left[\left(\frac{r_t - \mu_1}{\sqrt{\bar{\mu}_2}} \right)^4 \right]. \quad E(z^4)$$

- ▶ However, by itself it is not enough. Needs a comparison.

Compare with Normal dist.

Kurtosis

Since kurtosis deals with the mass away from the center of a distribution, then a “large” value of kurtosis should imply that the random variable has “tails” that are relatively thick.

Compare this number relative to a standard normal distribution.

Kurtosis of the standard normal random variable is exactly three.

Often represent the kurtosis through “excess kurtosis”:

$$\text{Ex. Kurt} = \bar{\mu}_4^s - 3.$$

1. Ex. Kurt < 0 , thinner tails than normal.
2. Ex. Kurt > 0 , thicker tails than normal.

Moment Estimators

- ▶ Clearly, we don't know the distribution of asset returns, or its population moments.
- ▶ However, we can estimate the “moments” of this distribution using sample data.
- ▶ There are common summary statistics that can be used to estimate $\bar{\mu}_1, \bar{\mu}_2, \bar{\mu}_3^s, \bar{\mu}_4^s$.

Expected Return

return $\Rightarrow$ %

- ▶ An estimator of an expectation is its sample average.

- ▶ So, how would we estimate expected return?

$$\bar{E} \Rightarrow \frac{1}{T} \sum_{t=1}^T$$

- ▶ Sample mean:

$$\bar{r} := \frac{1}{T} \sum_{t=1}^T r_t. \quad \%$$

- ▶ It is a good estimator of location.
- ▶ However, it is inappropriate if data are trending / non-stationary.

Uncertainty?

- ▶ Uncertainty is related to the potential fluctuations of an asset.
- ▶ So, how would we normally measure uncertainty?
- ▶ Sample variance:

$$s^2 := \frac{1}{T-1} \sum_{t=1}^T (r_t - \bar{r})^2. \quad \% ^2$$

- ▶ Estimator of uncertainty.

$$\begin{aligned} \text{cov}(x, x) &\Rightarrow \% ^2 \\ \underbrace{\text{cov}(x, x)}_{v(x)} \end{aligned}$$

Uncertainty?

- ▶ Sample standard deviation (std dev)

$$s := \sqrt{\frac{1}{T-1} \sum_{t=1}^T (r_t - \bar{r})^2}.$$

$\sqrt{\%^2} = \%$

- ▶ It is often used as a measure of the riskiness of an investment.
- ▶ Often called the volatility of a financial returns.
not variance.
std dev
- ▶ The standard deviation has the same scale as returns (the variance has the scale of returns squared) and is therefore easily interpretable.

Positive or Negative Extreme Returns?

- ▶ Need a measure of whether there are more **large** negative or positive returns.
- ▶ So, how would we normally measure this asymmetry?
- ▶ Sample Skewness

$$SK = \frac{1}{T} \sum_{t=1}^T \left(\frac{r_t - \bar{r}}{s} \right)^3 .$$

- ▶ Here, s stand for standard deviation where:

$$s^2 := \frac{1}{T-1} \sum_{t=1}^T (r_t - \bar{r})^2 .$$

Sample Skewness

- ▶ SK is a **sample estimate** of $\bar{\mu}_3^s$.
- ▶ If sample skewness is zero, the distribution is symmetric.
- ▶ If SK is negative, more likely to get **large negative** returns than positive.
- ▶ If SK is positive, more likely to obtain **large positive** returns than negative.
- ▶ Most stock returns exhibit negative skewness.

Overall Likelihood of Extreme Returns?

- ▶ Need a measure of likely extreme events (positive or negative).
- ▶ Sample Kurtosis

$$KT = \frac{1}{T} \sum_{t=1}^T \left(\frac{r_t - \bar{r}}{s} \right)^4$$

- ▶ KT meaningless in itself. Need a benchmark. We usually use the normal and often consider excess kurtosis.

$$\text{Ex. Kurt} = KT - 3$$

- ▶ If Ex. Kurt > 0 , tails are thicker than normal $\implies$ more extreme events than normal.
- ▶ If Ex. Kurt < 0 , tails are thinner than normal $\implies$ fewer extreme events than normal.

Microsoft stock and the S&P 500 index

We illustrate the descriptive statistical analysis of financial data using monthly closing prices on Microsoft stock and the S&P 500 index over the period January 1998 and May 2012.

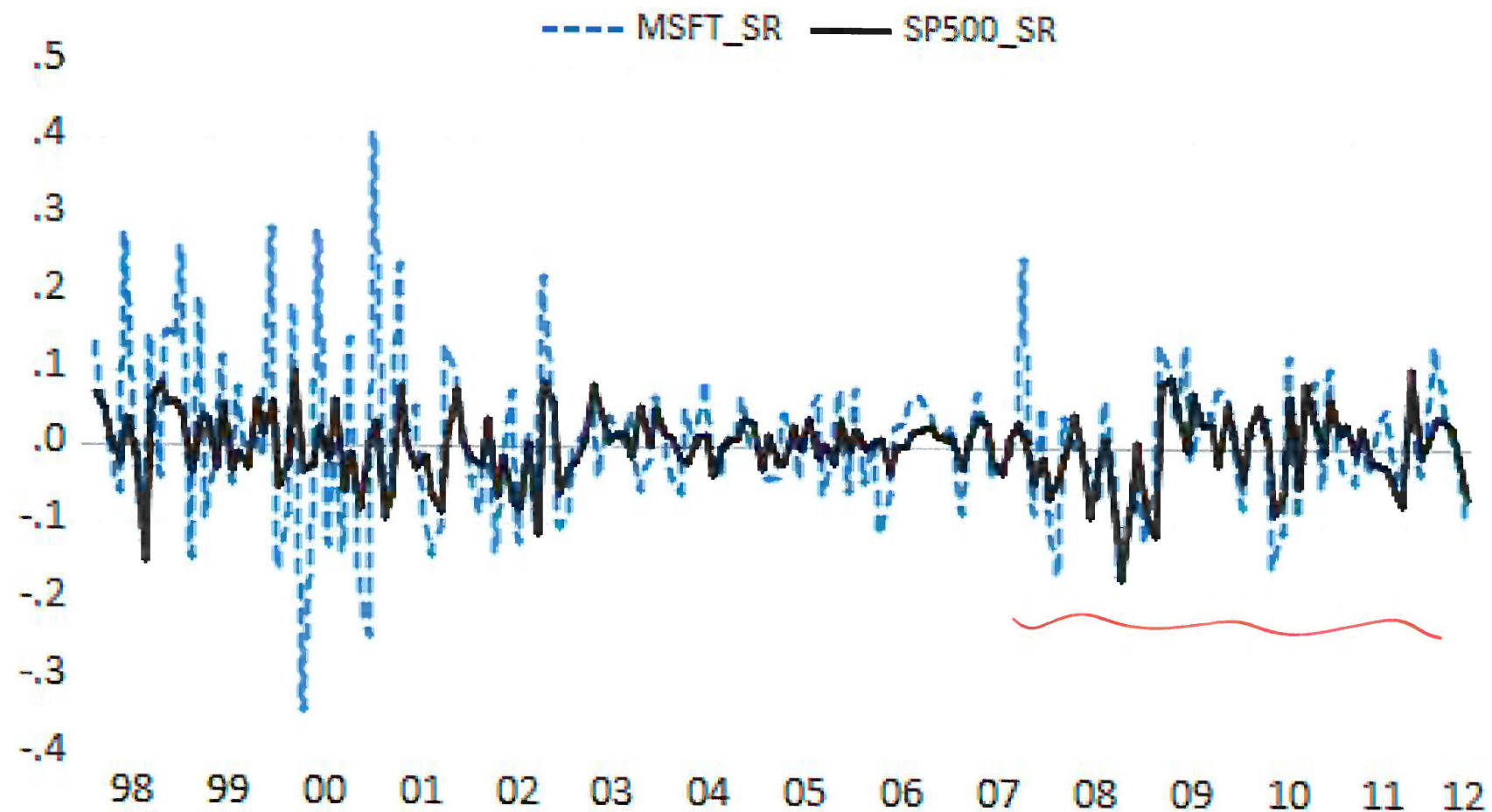
Prices on Microsoft stock, January 1998 to May 2012



S&P 500 index, January 1998 to May 2012



Monthly simple returns on Microsoft stock and the S&P 500 index.



Stationarity

The prices appear to be non-stationary since it has no tendency to revert to a time independent mean.

In contrast to prices, returns show clear mean-reverting behavior and the common monthly mean values look to be very close to zero.

Hence, the common mean value assumption of covariance stationarity looks to be satisfied.

The sample statistics for the Microsoft and S&P 500 monthly **simple** returns are given on the next slide.

Sample shape statistics

monthly

	MSFT	SP500
% ← Mean	0.0092	0.0028
% ² Variance	0.0103	0.0023
% ← Std Dev	0.1015	0.0478
Skewness	0.4853	−0.5631
Ex. Kurt	1.9148	0.6545

The mean and standard deviation for Microsoft monthly returns are 0.9% and 10%, respectively.

Annualized, these values are approximately 10.8% ($.009 \times 12$) and 34.6% ($.10 \times \sqrt{12}$), respectively.

The corresponding monthly and annualized values for S&P 500 returns are 0.3% and 4.7%, and 4.8% and 16.6%, respectively.

Sample shape statistics

high risk $\Rightarrow$ high return
No free lunch.

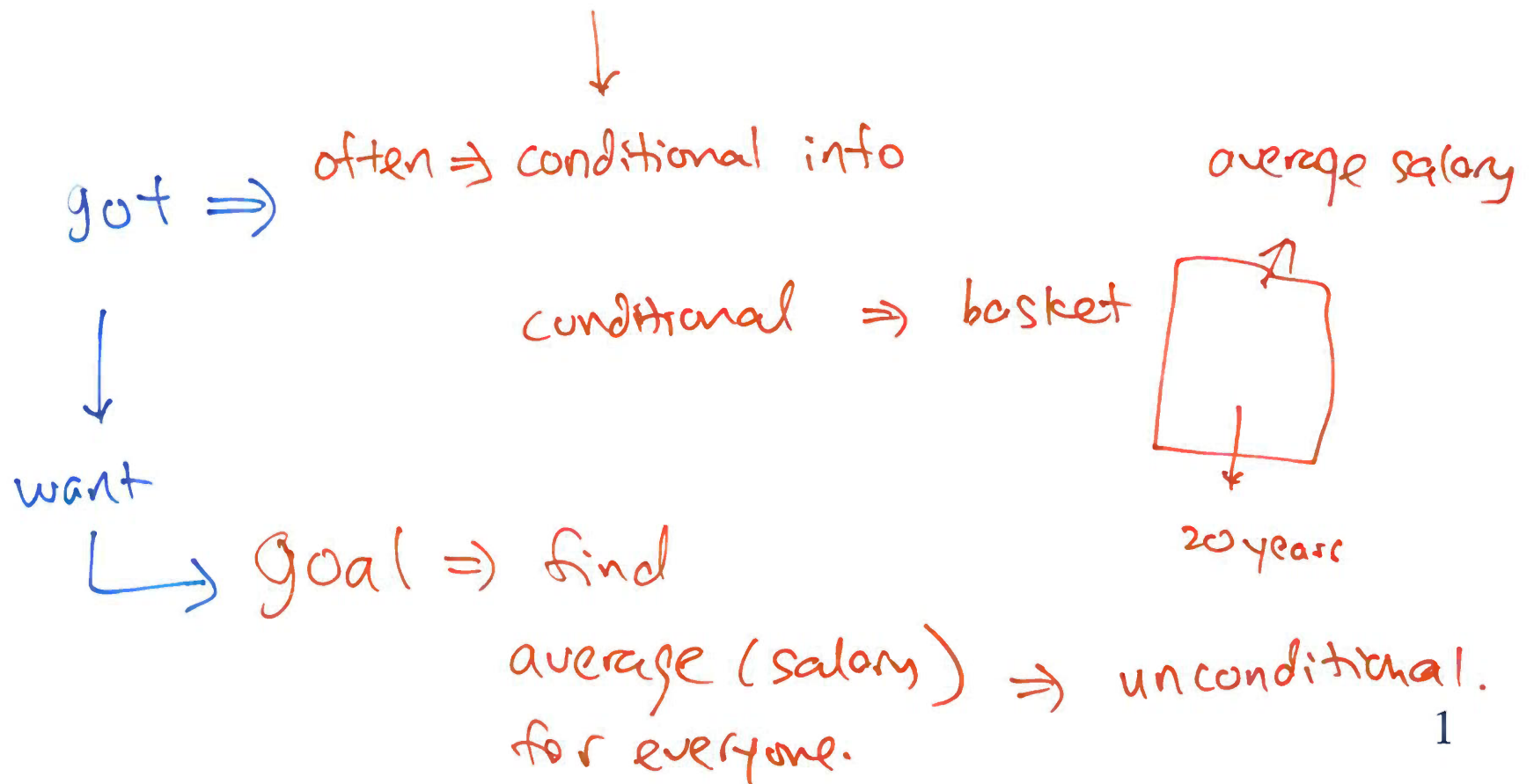
Microsoft has a higher mean and volatility than S&P 500.

The lower volatility for the S&P 500 reflects risk reduction due to diversification (holding many assets in a portfolio).

The sample skewness for Microsoft, 0.485, is slightly positive. The sample skewness for S&P 500, however, is moderately negative at -0.563, reflecting the somewhat long left tail of the histogram shown in Week 1.

The sample excess kurtosis values for Microsoft and S&P 500 are 1.915 and 0.655, respectively, and indicate that the tails are slightly fatter than the tails of a normal distribution.

Week 3: Co-movements of financial data



Multiple asset returns?

- ▶ When building portfolios, the investor wishes to know how assets move together.
- ▶ These features tell us about the potential risk in investing in an asset.
- ▶ How should we measure these?

→ only required info.

Joint Probability

then

Marginal Probability

← then

Conditional Probability

←

$$\Pr(X=1) = 0.605 = p$$

↓ ↓
R.V. small
 x

Joint Probability Distribution

Let X and Y be discrete random variables with sample spaces S_X and S_Y , respectively.

The likelihood that X and Y take values in the joint sample space $S_X \times S_Y$ is determined by the *joint probability distribution*

$$p(x, y) = \Pr(\underbrace{X = x, Y = y}_{\text{joint}}). = \text{a number}$$

$$0 \leq p(x, y) \leq 1$$

The function $p(x, y)$ satisfies:

$$\sum_{x \in S_X} \sum_{y \in S_Y} \Pr(X = x, Y = y) = 1.$$

^{toy} A simple example

Let E denote the performance of the economy next quarter.

For simplicity, say $E = 1$ if the economy is expanding and $E = 0$ if the economy is contracting.

E is a discrete random variable as we are able to list all the possible outcomes.

The sample space for E is $S_E = \{0, 1\}$.

Let S represent my sales for the next quarter, and assume the sample space for S is $S_S = \{0, 1, 2, 3\}$.

Table 1 (see the next slide) illustrates the joint distribution for S and E .



Table 1: Bivariate Distribution

each row

$S = 0$

s	recession 0	E 1 = e expansion	Pr(S)
0	1/8	0	1/8
1	2/8	1/8	3/8
2	1/8	2/8	3/8
3	0	1/8	1/8
marginal. $\Rightarrow$ Pr(E)	4/8	4/8	1

marginal prob. $\rightarrow$ table margin

across column

$$\Rightarrow P(S=0) = \frac{1}{8} + 0$$

$$Pr(S=0, E=0) = \frac{1}{8} \neq 0$$

Joint Probability Distribution

$$f(x) \equiv \text{pdf}$$

Notice that the total of all the entries in the table adds up to one.

each row $\Rightarrow$ sum across column

Sum $\rightarrow$ Sum

$$\sum_{s \in S_S} \sum_{e \in S_E} \Pr(S = s, E = e) = 1.$$

$\downarrow$ add inside the table.

0 1 2 3 0, 1

1 sum

The joint probability distribution tells us the probability of S and E occurring together.

What if we only want to know about the probability of S occurring, or the probability of E occurring?

x_1, x_2
continuous.

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x_1, x_2) \partial x_1 \partial x_2 = 1$$

joint pdf

$\Rightarrow$ formula $\Rightarrow$ slide 35.

slide 35

Marginal probability

$$\gamma = E$$

The probability $\Pr(X = x)$ is called the *marginal probability* of X and is given by:

$$\Pr(X = x) = \sum_{y \in S_Y} \Pr(X = x, Y = y).$$

each row $\Rightarrow$ across column.

$$\begin{array}{l} E=0 \\ E=1 \end{array}$$

Similarly, the *marginal probability* of $Y = y$ is given by:

$$\Pr(Y = y) = \sum_{x \in S_X} \Pr(X = x, Y = y).$$

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X_1 and X_2
continuous

$$f_1(x_1) = \int_{-\infty}^{\infty} f(x_1, x_2) dx_2$$

↓
marginal pdf

sum across column

↓
 y

Find $\Pr(S = 0)$ and $\Pr(E = 1)$ from the joint distribution in Table 1.

What is $\Pr(S = 0)$ regardless of the value of E ?

each row = across column.

$$\begin{aligned}\Pr(S = 0) &= \Pr(S = 0, E = 0) + \Pr(S = 0, E = 1) \\ &= \frac{1}{8} + 0 \\ &= \frac{1}{8}\end{aligned}$$

Notice that this probability is equal to the horizontal (row) sum of the probabilities in Table 1 at $S = 0$.

Marginal Distributions

Similarly, we can find $\Pr(E = 1)$ as follows:

each column = ^{Sum} across row

$$\begin{aligned}\Pr(E = 1) &= \Pr(S = 0, E = 1) + \Pr(S = 1, E = 1) + \\ &\quad \Pr(S = 2, E = 1) + \Pr(S = 3, E = 1) \\ &= 0 + \frac{1}{8} + \frac{2}{8} + \frac{1}{8} \\ &= \frac{4}{8}\end{aligned}$$

This probability is the vertical (column) sum of the probabilities in Table 1 at $E = 1$.

Marginal Distributions

Why we call *marginals* probability “marginals” ?

The table represents the joint and in the margins, we get the marginals.

Conditional Distributions

The conditional probability that $X = x$ given that $Y = y$ (provided $\Pr(Y = y) \neq 0$) is,

$$p(x|y) = \Pr(X = x|Y = y) = \frac{\Pr(X = x, Y = y)}{\Pr(Y = y)} = \frac{\text{Joint}}{\text{marginal.}}$$

given that we know $Y=y$ thus we know

The conditional probability that $Y = y$ given that $X = x$ (provided $\Pr(X = x) \neq 0$) is,

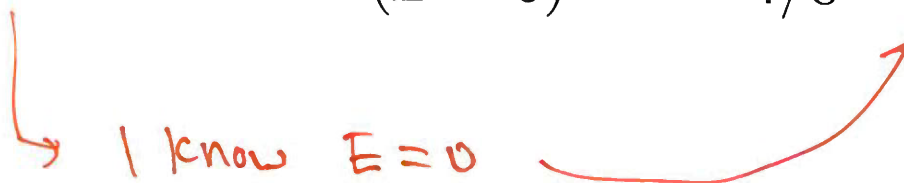
$$p(y|x) = \Pr(Y = y|X = x) = \frac{\Pr(X = x, Y = y)}{\Pr(X = x)}$$

Conditional Distributions

What is the probability that $S = 0$ given that we know $E = 0$?

To find this probability, we compute the conditional probability:

$$\Pr(S = 0|E = 0) = \frac{\Pr(S = 0, E = 0)}{\Pr(E = 0)} = \frac{1/8}{4/8} = \frac{1}{4}.$$

 I know $E = 0$

The notation $\Pr(S = 0|E = 0)$ is read as “the probability that $S = 0$ given that $E = 0$ ”.

Conditional Distributions

Notice that

$$\Pr(S = 0|E = 0) = \frac{1}{4} > \Pr(S = 0) = \frac{1}{8}.$$

Handwritten annotations:

- conditional (with an arrow pointing to the conditional probability term)
- marginal (with an arrow pointing to the marginal probability term)
- not knowing E (with an arrow pointing to the marginal probability term)
- not equal. (with an arrow pointing to the inequality symbol)

Hence, knowledge that $E = 0$ increases the likelihood that $S = 0$.

Clearly, S depends on E .

knowing E is very useful

Table 2: Conditional distributions

Knowing $E=0$ ↓		Knowing $E=1$ ↗	
s	$\Pr(S E=0)$	s	$\Pr(S E=1)$
0	2/8	0	0
1	4/8	1	2/8
2	2/8	2	4/8
3	0	3	2/8
sum 1		sum 1	

The first column is the conditional distribution of S given $E = 0$.

The second column is the conditional distribution of S given $E = 1$.

We read: the probability of Sales of 3 ($S = 3$) given (or conditional on) the economy is growing ($E = 1$) is $2/8$.

Discrete random variables

In general the notation is...

- ① ► $\Pr(Y = y, X = x)$ is the **joint probability** of the random variable Y equal y **AND** the random variable X equal x . *only info required*
- ② ► $\Pr(Y = y)$ and $\Pr(X = x)$ are the **marginal probabilities** of $Y = y$ and $X = x$.
Then
- $\Pr(Y = y \mid X = x)$ is the **conditional probability** of the random variable Y takes the value y **GIVEN** that X equals x . $\Rightarrow \frac{\text{①}}{\text{②}}$

Independence

1:00

1:05 back

Intuitively, X is independent of Y if knowledge about Y does not influence the likelihood that $X = x$.

We represent this intuition formally as follows.

Let X and Y be discrete random variables with sample spaces S_X and S_Y , respectively. X and Y are independent random variables if and only if:

① $\Pr(X = x | Y = y) = \Pr(X = x), \quad \text{for all } x \in S_X, y \in S_Y$

conditional = marginal

Y is useless.

② $\Pr(X = x | Y = y) = \frac{\Pr(X = x, Y = y)}{\Pr(Y = y)} = \Pr(X = x)$

joint = marginal

$= \frac{\Pr(X = x) \Pr(Y = y)}{\Pr(Y = y)} = \Pr(X = x)$

joint marginal

Independence

slide 15

For the data in Tables 1 and 2, we know that

$$\Pr(S = 0|E = 0) = \frac{1}{4} \neq \Pr(S = 0) = \frac{1}{8}.$$

Thus S and E are not independent.

E is a very useful info.

Expected Value and Conditional Expectation

↓
unconditional vs ↗

Expected Value $E(RV) = \text{number} \times \text{Pr} + \text{number} \times \text{Pr}$

For a discrete random variable X with sample space S_X , the expected value is defined as:

$$\mathbb{E}[X] = \sum_{x \in S_X} x \cdot \text{Pr}(X = x).$$

We weight each possible value by how likely they are. . .

This provides us with a measure of centrality of the distribution.

A “good” prediction for X !

Table 1: Bivariate Distribution slide 6

s	0	1	$\Pr(S)$
0	$1/8$	0	$1/8$
1	$2/8$	$1/8$	$3/8$
2	$1/8$	$2/8$	$3/8$
3	0	$1/8$	$1/8$
$\Pr(E)$	$4/8$	$4/8$	1

Expected Value

Using the discrete distribution in Table 1, the expected return is computed as:

$$\mathbb{E}[S] = \overset{S}{0} \cdot \underset{Pr}{(1/8)} + \overset{S}{1} \cdot \underset{Pr}{(3/8)} + \overset{S}{2} \cdot (3/8) + \overset{S}{3} \cdot (1/8)$$

$$= 1.5$$

 I don't care about $\mathbb{E}$

Conditional Expectation

$\mathbb{E}[X|Y = y]$ is the conditional mean of $X|Y = y$.

For discrete random variables X and Y , the conditional expectations are defined as:

$$\mathbb{E}[X|Y = y] = \sum_{x \in S_X} x \cdot \underbrace{\Pr(X = x|Y = y)}_{\text{cond. prob.}},$$

$$\mathbb{E}[Y|X = x] = \sum_{y \in S_Y} y \cdot \Pr(Y = y|X = x).$$

Table 2: Conditional distributions (duplicate)

s	$\Pr(S E = 0)$		s	$\Pr(S E = 1)$	
0	x 2/8	+	0	x 0	+
1	x 4/8	+	1	x 2/8	+
2	2/8		2	4/8	
3	0		3	2/8	

Tell me $\Rightarrow E(S|E=0) = 1$ and $E(S|E=1) = 2$

get

$\xrightarrow{\text{LIE}} E(S) = 1.5 \Rightarrow \text{"a number" not R.V.}$

$\Rightarrow$ conditional info

$\downarrow$ LIE

unconditional

Conditional Expectation

$E(S|E)$ is a R.V.
 $E(S)$ is not a R.V.

For the random variables in Table 2 (Conditional Distributions), we have the following conditional moments for S :

$$\mathbb{E}[S|E = 0] = 0 \cdot \frac{1}{4} + 1 \cdot \frac{1}{2} + 2 \cdot \frac{1}{4} + 3 \cdot 0 = 1 \quad \text{tail}$$

OR

$$\mathbb{E}[S|E = 1] = 0 \cdot 0 + 1 \cdot \frac{1}{4} + 2 \cdot \frac{1}{2} + 3 \cdot \frac{1}{4} = 2 \quad \text{head}$$

Before E is observed, $\mathbb{E}[S|E]$ is a random variable because its value depends on the value of E .

Since $\mathbb{E}[S|E]$ is a random variable, we can consider their mean values.

To illustrate: number x Pr + number x Pr.

$$\overset{\text{tail}}{E[S|E=0]} \times \text{Pr}(E=0) + \overset{\text{head}}{E[S|E=1]} \times \text{Pr}(E=1)$$

$$= 1 \times \frac{1}{2} + 2 \times \frac{1}{2}$$

$$= 1.5$$

$$= E(S)$$

This result is known as the **Law of Iterated Expectations (LIE)**.

$$E(S) = E(E(S|E)) \Rightarrow \text{L.I.E}$$

average across head and tail E=0 or E=1

Law of Iterated Expectations

For two random variables X_1 and X_2 (discrete or continuous), the Law of Iterated Expectations (LIE) states that

$$\mathbb{E}[X_1] = \mathbb{E}[\mathbb{E}[X_1|X_2]]. \quad \checkmark \quad \text{LIE}$$

In words, the LIE says that the mean of the conditional mean is the unconditional mean.

True or false

$$\mathbb{E}(X_1) \stackrel{??}{=} \mathbb{E} \left(\mathbb{E}(X_1 \mid \overset{\text{big}}{\downarrow} X_2 = \overset{\text{small}}{\swarrow} x_2) \right) \quad \text{False.}$$

$\swarrow$
outer

$\downarrow$
not a particular outcome
for X_2

Intuition: LIE



1. The inner expectation $E[X_1|X_2]$ represents the conditional expectation of X_1 given X_2 . It treats X_2 as known.
2. The outer expectation $E[\cdot]$ then takes the expectation of that conditional expectation over all possible values of X_2 .



$E=1$ and $E=0$

Why is it useful?

It simplifies calculations when direct computation of $E[X_1]$ is difficult.
It is widely used in finance and econometrics.

Conditional variance

$$\text{var}(X) = E(X - E(X))^2 =$$

Not R.V.
a number
with no
prob.

A very useful conditional moment is the conditional variance.

Definition

The variance of a random variable Y , conditional on X is given by

$$\text{Var}(Y|X) = E[(Y - E[Y|X])^2 | X] . \Rightarrow \text{R.V.}$$

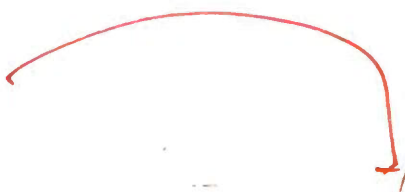

Law of Total Variance:

The variance of Y can be equivalently calculated as

$$\text{Var}(Y) = E[\underbrace{\text{Var}(Y|X)}_{\text{R.V.}}] + \text{Var}(\underbrace{E[Y|X]}_{\text{R.V.}}).$$

↓
unconditional

→ info about $V(E(Y|X)) + E(\text{var}(Y|X))$



Continuous Multivariate Random Variables

① $f(x_1, x_2) \Rightarrow$ Joint $\Rightarrow$ formula
p.d.f. Joint

Then

② $f_1(x_1)$ and $f_2(x_2)$

Then

③ $f(x_1 | x_2) = \frac{\text{Joint}}{\text{marginal.}}$

Joint Distributions

Multivariate random variables allow relationship between two or more random quantities to be modelled and studied.

We will use bold face type to indicate a column vector with n components

$$\mathbf{X} = \begin{pmatrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{pmatrix}$$

return Asset 1

return Asset n

Each of the X_i s is a random variable.

Joint Distributions

- ▶ Multivariate random variables can be either discrete or continuous.
- ▶ Discrete multivariate random variables are fairly uncommon in financial econometrics and so we will focus on the continuous case.
- ▶ The joint distribution of multivariate random variables can be easily understood in the bivariate case.
- ▶ Consider the bivariate case: $\mathbf{X} = (X_1, X_2)'$.

Joint Distributions

Let $\mathbf{X} = (X_1, X_2)'$ be continuous random variables defined over the real line.

We characterize the joint probability distribution of X_1 and X_2 using the joint probability density function (pdf) $f(x_1, x_2)$ such that $f(x_1, x_2) \geq 0$ and,

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x_1, x_2) dx_1 dx_2 = 1.$$

sum
inside
the
table

slide 7

Marginal Distribution

Let $\mathbf{X} = (X_1, X_2)'$ with joint pdf $f(x_1, x_2)$.

The marginal pdf of X_1 is given by

$$f_1(x_1) = \int_{-\infty}^{\infty} f(x_1, x_2) dx_2.$$

each row = across column "sum"

Likewise, the marginal pdf of X_2 is given by

$$f_2(x_2) = \int_{-\infty}^{\infty} f(x_1, x_2) dx_1.$$

each column = across row

The marginal distribution requires integrating out the variables which are not of interest.

If X_1 and X_2 are **jointly** normally distributed, then their joint behaviour is completely characterised by the mean vector and covariance matrix:

$$\begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \sim N \left(\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \begin{pmatrix} \sigma_{11}^2 & \sigma_{12} \\ \sigma_{12} & \sigma_{22}^2 \end{pmatrix} \right)$$

↓
mean
(1)
(3) Covariance
variance (2)

where $\sigma_{12} = \text{Cov}(X_1, X_2)$.

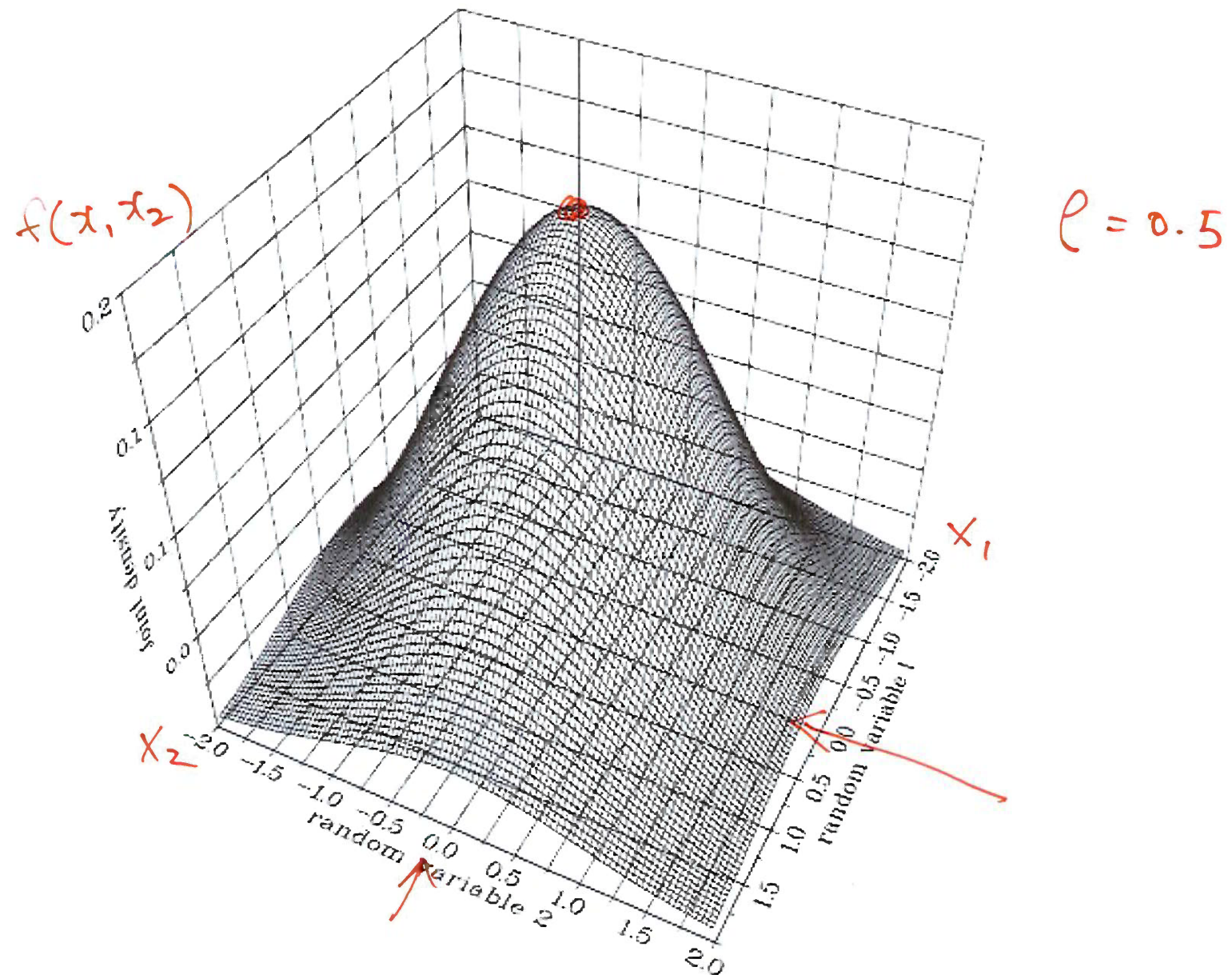
The **joint pdf** is given by

$$f(x_1, x_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} e^{-\frac{1}{2\sqrt{1-\rho^2}} \left[\frac{(x_1-\mu_1)^2}{\sigma_1^2} + \frac{(x_2-\mu_2)^2}{\sigma_2^2} - \frac{2\rho(x_1-\mu_1)(x_2-\mu_2)}{\sigma_1\sigma_2} \right]}$$

where

$$-1 \leq \rho = \frac{\sigma_{12}}{\sigma_1\sigma_2} \leq 1$$

Joint density for two standard normally distributed random variables and $\rho = 0.5$.



Conditional Distributions

- ▶ Our goal will often be to understand the behavior of X_1 , conditional on X_2 , taking a specific value $X_2 = x_2$ or within some range of values.
- ▶ Vast majority of financial econometrics is interested in understanding the behavior of a random variable conditional on the observed values of some other random variables.
- ▶ We define the conditional distribution of a random variable, or its moments, by making use of the joint and marginal distributions.
- ▶ Consider $\mathbf{X} = (X_1, X_2)'$.

Conditional Distribution

Let $\mathbf{X} = (X_1, X_2)'$ be a bivariate random variable with joint pdf $f(x_1, x_2)$ and marginal densities $f_1(x_1)$ for X_1 , and $f_2(x_2)$ for X_2 .

The **conditional pdf** of X_1 given $X_2 = x_2$, is

$$f(x_1|x_2) = \frac{f(x_1, x_2)}{f_2(x_2)}. \quad = \quad \frac{\text{Joint}}{\text{marginal}}$$

Independence

Let $\mathbf{X} = (X_1, X_2)'$ with joint pdf $f(x_1, x_2)$ and marginal pdfs $f_1(x_1)$, $f_2(x_2)$.

The random variables X_1 and X_2 are independent if and only if

$$f(x_1, x_2) = f_1(x_1)f_2(x_2).$$

$$\text{Joint} = \text{marginal} \underline{\underline{=}}$$

This result is extremely useful in practice because it gives us an easy way to compute the joint pdf for two independent random variables: we simply compute the product of the marginal distributions.

$$f(x_1 | x_2) = f_1(x_1)$$

↓
useless.

Independence

Let X_1 and X_2 be continuous random variables with marginal pdfs $f_1(x_1)$ and $f_2(x_2)$, and joint pdf $f(x_1, x_2)$.

Then X_1 and X_2 are independent if and only if

$$f(x_1|x_2) = \frac{f(x_1, x_2)}{f_2(x_2)} = \frac{f_1(x_1)f_2(x_2)}{f_2(x_2)} = f_1(x_1).$$

That is, if variables are independent, the conditional distribution is nothing but the marginal distribution.